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Geared ramp assembly with raised ramp position and side rails and method of use

Abstract

A ramp assembly to accommodate a wheel-chaired passenger to enter or exit a motorized vehicle having a conventional floor includes a frame, a ramp, and a side rail assembly. The ramp is slidably coupled to the frame for movement between a stowed position and an extended position. Once extended, the rear end of the ramp is movable between a raised position and a lowered position. The side rail assembly is moveable from a collapsed position to a lifted position as the ramp moves from the lowered position to the raised position.

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Background/Summary

CROSS REFERENCE TO OTHER APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 18/082,198, filed Dec. 15, 2022, entitled “Geared Ramp Assembly with Raised Ramp Position and Side Rails and Method of Use,” which is a continuation of U.S. patent application Ser. No. 17/094,930, filed Nov. 11, 2020, entitled “Geared Ramp Assembly with Raised Ramp Position and Side Rails and Method of Use,” which claims the benefit of U.S. Provisional Patent Application Ser. No. 63/026,777, filed May 19, 2020, entitled “Geared Ramp Assembly with Raised Ramp Position and Side Rails and Method of Use”, and U.S. Provisional Patent Application Ser. No. 62/945,291, filed Dec. 9, 2019, entitled “Geared Ramp Assembly with Raised Ramp Position and Method of Use”, the disclosures of which are hereby incorporated by reference in their entirety.

FIELD OF THE DISCLOSURE

(1) The present application relates to a passenger vehicle for transporting one or more passengers, and more particularly to a ramp assembly movable to a raised position for accommodating ingress and egress of a physically limited passenger.

BACKGROUND

(2) Automobile manufacturers do not currently mass-produce passenger motor vehicles specifically designed to transport passengers having physical limitations, either as a driver or as a non-driving passenger. Consequently, mass-produced passenger vehicles are modified, or retrofitted, by a number of aftermarket companies dedicated to supplying vehicles to physically limited passengers. Such vehicles can be modified altering or adding certain parts or structures within a vehicle to accommodate the physically limited passenger without inconveniencing other passengers or sacrificing space in the vehicle. For example, in one configuration, a van or bus is retrofitted with a ramp to enable a physically limited individual using a wheelchair to enter and exit the vehicle.

(3) In some instances, the ramp is stored below the conventional vehicle floor and deploys to accommodate entry and exit of the physically limited individual through a side door or entrance of the vehicle. Challenges may arise related to stowing the ramp and ensuring that the ramp is flush with the conventional vehicle floor when the ramp is deployed.

SUMMARY OF THE EMBODIMENTS

(4) In a first embodiment of the disclosure, a ramp assembly to accommodate a wheel-chaired passenger to enter or exit a motorized vehicle having a conventional floor includes a ramp, a drive block, and a wheel. The frame includes a top edge, a bottom edge, a front end, a rear end spaced longitudinally apart from the front end, and a track system extending longitudinally between the front end and the rear end of the frame. The ramp is slidably coupled to the frame and includes a front end and a rear end. The drive block is configured to translate longitudinally along the track system. The wheel is pivotally coupled to the ramp, has a longitudinal axis, and is rotatably coupled to the drive block for rotation about the longitudinal axis.

(5) In some embodiments, the ramp is configured to raise and lower relative to the wheel when the wheel is rotated about the longitudinal axis. The ramp assembly may further include a hinge having a first end coupled to the wheel and a second end coupled to the rear end of the ramp, wherein the first end is configured to pivot relative to the second end to facilitate pivoting movement of the ramp relative to the wheel.

(6) In some embodiments, the wheel is rotatable between a first position in which the rear end of the ramp is positioned below the top edge of the frame, and a second position in which the rear end of the ramp is positioned above the top edge of the frame. In some embodiments, the ramp assembly further comprises a toothed-mount coupled to the front end of the frame, wherein the wheel is a toothed-wheel configured to rotate about the longitudinal axis when the toothed-wheel is in contact with the toothed-mount.

(7) In some embodiments, the toothed-wheel is rotatable between a first position in which the rear end of the ramp is positioned below the toothed-mount, and a second position in which the rear end

of the ramp is positioned above the toothed-mount. The ramp is movable between: (i) a stowed position in which the wheel is spaced apart from the front end of the frame, (ii) a lowered position in which the wheel is positioned at the front end of the frame, (iii) and a raised position in which the wheel is positioned at the front end of the frame and rotated relative to the wheel when the ramp is in the lowered position. In the stowed position the ramp is positioned inside the motorized vehicle and beneath the conventional floor. In the lowered position the ramp is positioned outside the motorized vehicle and the rear end of the ramp is positioned below the conventional floor. In the raised position the ramp is positioned outside the motorized vehicle and the rear end of the ramp is positioned flush with the conventional floor.

(8) In some embodiments, the track system includes: a first-side track, a second-side track spaced apart from the first-side track, and an opening defined between the first-side track and the second-side track sized to receive the ramp. In some embodiments, the ramp assembly includes a mount coupled to the first-side track and the second-side track, wherein the mount further defines the opening sized to receive the ramp. In some embodiments, the ramp assembly further includes a first bearing coupled to the mount; a second bearing positioned at the rear end of the frame; a threaded-rod extending between and configured to rotate relative to the first bearing and the second bearing; and a threaded-casing fixedly coupled to the drive block and configured to translate longitudinally along the threaded-rod during rotation of the threaded-rod. In some embodiments, the wherein the wheel is rotatable between a first position in which the rear end of the ramp is aligned with the opening, and a second position in which the rear end of the ramp is positioned above the mount.

(9) In another embodiment of the disclosure, a ramp assembly for a motorized vehicle having a conventional floor includes a frame and a ramp. The frame includes a front end, a rear end spaced longitudinally apart from the front end, and a track system extending longitudinally between the front end and the rear end of the frame. The ramp is slidably coupled to the frame and includes a front end and a rear end. The ramp assembly further includes a drive block configured to translate longitudinally along the track system; a first cog coupled to the frame; a second cog coupled to the drive block for longitudinal movement therewith. One of the first cog and the second cog is a toothed-wheel pivotally coupled to the ramp, and the other of the first cog and the second cog is a toothed-bar configured to engage the toothed-wheel to cause rotation of the toothed-wheel.

(10) In some embodiments, the ramp is configured to pivot relative to the toothed-wheel during rotation of the toothed-wheel. In some embodiments, the track system includes a first-side track and a second-side track, and the toothed-wheel is rotatable between: a first position in which the rear end of the ramp is positioned above the frame, and a second position in which the rear end of the ramp is vertically aligned with an opening of the frame defined between the first-side track and the second-side track.

(11) In some embodiments, the ramp is movable between: (i) a stowed position in which the second cog is spaced apart from the first cog, (ii) a lowered position in which the second cog is in contact with the first cog, (iii) and a raised position in which the second cog is in contact with the first cog and the toothed-wheel is rotated relative to the toothed-wheel when the ramp is in the lowered position. In the stowed position the ramp is positioned inside the motorized vehicle and beneath the conventional floor. In the lowered position the rear end of the ramp is positioned below the conventional floor. In the raised position the ramp is positioned outside the motorized vehicle and the rear end of the ramp is positioned flush with the conventional floor.

(12) In some embodiments, the ramp is movable between: (i) a stowed position in which the second cog is spaced apart from the rear end of the frame, (ii) a lowered position in which the second is positioned at the rear end of the frame, (iii) and a raised position in which the second cog is positioned at the rear end of the frame and the toothed-wheel is rotated relative to the toothed-wheel when the ramp is in the lowered position.

(13) In another embodiment of the disclosure, a method of moving a ramp relative to a conventional floor of a motorized vehicle includes: sliding a rear end of the ramp from a rear end to

a front end of a frame that is positioned beneath the conventional floor; and pivoting the rear end of the ramp relative to the frame from first position in which the rear end of the ramp is below the conventional floor to a second position in which the rear end of the ramp is flush with the conventional floor.

(14) In some embodiments, sliding a rear end of the ramp from a rear end to a front end of a frame includes engaging a first cog with a second cog. In some embodiments, pivoting the rear end of the ramp relative to the frame includes rotating the one of the first cog and second cog relative to the other of the first cog and second cog.

(15) In a further embodiment of the present disclosure, a ramp assembly is configured to accommodate a wheel-chaired passenger to enter or exit a motorized vehicle. The motorized vehicle has a conventional floor. The ramp assembly comprises: a frame including a top edge, a bottom edge, a front end, a rear end spaced longitudinally apart from the front end, and a track system extending longitudinally between the front end and the rear end of the frame; a ramp slidably coupled to the frame and including a front end and a rear end; a drive block configured to move longitudinally along the track system; and a wheel pivotally coupled to the ramp; wherein the wheel has a longitudinal axis and is rotatably coupled to the drive block for rotation about the longitudinal axis.

(16) In some embodiments, the ramp is configured to be raised and lowered relative to the wheel when the wheel is rotated about the longitudinal axis. In some embodiments, the ramp assembly includes a hinge having a first end coupled to the wheel and a second end coupled to the rear end of the ramp, wherein the first end of the hinge is configured to pivot relative to the second end of the hinge to facilitate pivoting movement of the ramp relative to the wheel.

(17) In some embodiments, the wheel is rotatable between: a first position in which the rear end of the ramp is positioned below the top edge of the frame, and a second position in which the rear end of the ramp is positioned above the top edge of the frame.

(18) In some embodiments, the ramp assembly further comprises a cross bar coupled to the front end of the frame, wherein the cross bar includes a toothed portion; wherein the wheel is a toothed wheel configured to rotate about the longitudinal axis when in contact with the toothed portion of the cross bar. In some embodiments, the toothed wheel is rotatable between: a first position in which the rear end of the ramp is positioned below the toothed mount, and a second position in which the rear end of the ramp is positioned above the toothed mount.

(19) In some embodiments, the ramp is movable between: (i) a stowed position in which the wheel is spaced apart from the front end of the frame, (ii) a lowered position in which the wheel is positioned at the front end of the frame, (iii) and a raised position in which the wheel is positioned at the front end of the frame and rotated relative to the wheel when the ramp is in the lowered position.

(20) In some embodiments, in the stowed position the ramp is positioned inside the motorized vehicle and beneath the conventional floor, in the lowered position the ramp is positioned outside the motorized vehicle and the rear end of the ramp is positioned below the conventional floor, and in the raised position the ramp is positioned outside the motorized vehicle and the rear end of the ramp is positioned flush with the conventional floor.

(21) In some embodiments, the track system includes: a first-side track coupled a first side of the frame, a second-side track spaced apart from the first-side track and coupled to a second side of the frame, and an opening defined between the first-side track and the second-side track sized to receive the ramp. In some embodiments, the ramp assembly includes a cross bar coupled to the first side of the frame and the second side of the frame, wherein the cross bar further defines the opening sized to receive the ramp.

(22) In some embodiments the ramp assembly comprises: a first bearing coupled to the cross bar; a second bearing positioned at the rear end of the frame; a threaded rod extending between and configured to rotate relative to the first bearing and the second bearing; and a threaded casing fixed

relative to the drive block and configured to move longitudinally along the threaded rod during rotation of the threaded rod. In some embodiments, the wheel is rotatable between: a first position in which the rear end of the ramp is aligned with the opening, and a second position in which the rear end of the ramp is positioned above the cross bar.

(23) In another illustrative embodiment, a ramp assembly for a motorized vehicle having a conventional floor comprises: a frame including a front end, a rear end spaced longitudinally apart from the front end, and a track system extending longitudinally between the front end and the rear end of the frame; a ramp slidably coupled to the frame and including a front end and a rear end; a drive block configured to translate longitudinally along the track system; a first cog coupled to the frame; and a second cog coupled to the drive block for longitudinal movement therewith, wherein one of the first cog and the second cog is a toothed wheel pivotally coupled to the ramp, and the other of the first cog and the second cog is a toothed bar configured to engage the toothed wheel to cause rotation of the toothed wheel.

(24) In some embodiments, the ramp is configured to be raised and lowered relative to the toothed wheel during rotation of the toothed wheel. In some embodiments, the track system includes a first-side track and a second-side track, and the toothed wheel is rotatable between: a first position in which the rear end of the ramp is positioned above the frame, and a second position in which the rear end of the ramp is vertically aligned with an opening of the frame defined between the first-side track and the second-side track.

(25) In some embodiments, the ramp is movable between: (i) a stowed position in which the second cog is spaced apart from the first cog, (ii) a lowered position in which the second cog is in contact with the first cog, (iii) and a raised position in which the second cog is in contact with the first cog and the toothed wheel is rotated relative to the toothed wheel when the ramp is in the lowered position.

(26) In some embodiments, in the stowed position the ramp is positioned inside the motorized vehicle and beneath the conventional floor, in the lowered position the rear end of the ramp is positioned below the conventional floor, and in the raised position the ramp is positioned outside the motorized vehicle and the rear end of the ramp is positioned flush with the conventional floor. In some embodiments, the ramp is movable between: (i) a stowed position in which the second cog is spaced apart from the rear end of the frame, (ii) a lowered position in which the second is positioned at the rear end of the frame, (iii) and a raised position in which the second cog is positioned at the rear end of the frame and the toothed wheel is rotated relative to the toothed wheel when the ramp is in the lowered position.

(27) In another illustrative embodiment, a method of moving a ramp relative to a conventional floor of a motorized vehicle comprises: sliding a rear end of the ramp from a rear end to a front end of a frame that is positioned beneath the conventional floor; and adjusting the height of the rear end of the ramp relative to the frame from first position in which the rear end of the ramp is below the conventional floor to a second position in which the rear end of the ramp is flush with the conventional floor.

(28) In some embodiments, sliding a rear end of the ramp from a rear end to a front end of a frame includes engaging a first cog with a second cog; and adjusting the height of the rear end of the ramp relative to the frame includes rotating the one of the first cog and second cog relative to the other of the first cog and second cog.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The above-mentioned aspects of the present disclosure and the manner of obtaining them will become more apparent and the disclosure itself will be better understood by reference to the

following description of the embodiments of the disclosure, taken in conjunction with the accompanying drawings, wherein:

- (2) FIG. 1 is a perspective view of a passenger vehicle including a movable ramp;
- (3) FIG. 2A is a perspective view of a ramp assembly showing a ramp in a stowed position beneath a conventional floor of the passenger vehicle;
- (4) FIG. 2B is a partial cross-section view of the ramp assembly of FIG. 2A showing the ramp engaged with a track system of a frame;
- (5) FIG. 3A is a perspective view of the ramp assembly showing the ramp in a lowered position below the conventional floor of the passenger vehicle;
- (6) FIG. 3B is a partial cross-section view of the ramp assembly of FIG. 3A showing a toothed-wheel in contact with a toothed-bar of the frame;
- (7) FIG. 4A is a perspective view of the ramp assembly showing the ramp in a raised position flush with the conventional floor of the passenger vehicle;
- (8) FIG. 4B is a partial cross-section view of the ramp assembly of FIG. 4A showing the toothed-wheel rotated relative to the toothed-wheel in FIG. 3B;
- (9) FIG. 5 is a perspective view of a passenger vehicle including a movable ramp;
- (10) FIG. 6A is a perspective view of a ramp assembly showing a ramp in a stowed position beneath a conventional floor of the passenger vehicle;
- (11) FIG. 6B is a partial cross-section view of the ramp assembly of FIG. 6A showing the ramp engaged with a track system of a frame;
- (12) FIG. 6C is a cut away perspective view of cogs of a ramp assembly including a toothed bar and a toothed wheel;
- (13) FIG. 7A is a perspective view of the ramp assembly showing the ramp in a lowered position below the conventional floor of the passenger vehicle;
- (14) FIG. 7B is a partial cross-section view of the ramp assembly of FIG. 7A showing a toothed-wheel in contact with a toothed-bar of the frame;
- (15) FIG. 8A is a perspective view of the ramp assembly showing the ramp in a raised position flush with the conventional floor of the passenger vehicle;
- (16) FIG. 8B is a partial cross-section view of the ramp assembly of FIG. 8A showing the toothed wheel rotated relative to the toothed wheel in FIG. 7B;
- (17) FIG. 9 is a partial cross-section view of a side rail of the ramp assembly showing the side rail in a collapsed position; and
- (18) FIG. 10 is a partial cross-section view of the side rail of the ramp assembly showing the side rail in a lifted position.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(19) The embodiments of the present disclosure described below are not intended to be exhaustive or to limit the disclosure to the precise forms disclosed in the following detailed description. Rather, the embodiments are chosen and described so that others skilled in the art may appreciate and understand the principles and practices of the present disclosure.

(20) FIG. 1 illustrates a vehicle **100**, commonly identified as a passenger van or bus, available from any number of United States and foreign manufacturers. The vehicle may be a single-level bus, a double-decker bus, or any other type of vehicle. The principles and teachings of the present disclosure may be used for any type of vehicle.

(21) In the illustrative embodiment shown in FIG. 1, the vehicle **100** includes a unibody construction, but other vehicles having a frame on body construction are also included in the present disclosure. Consequently, the use of vehicle herein includes all types and kinds of vehicles with a body on frame construction, a unibody construction, or other constructions. In addition, while the passenger van **100** is illustrated in FIG. 1, the present disclosure is directed to all passenger vehicles carrying one or more passengers including, but not limited to, a bus, motor coach, sport-utility vehicle, truck, taxi, ambulance, or passenger car.

(22) The vehicle **100** includes a frame or chassis **102** operatively coupled to a first pair of wheels **104** and a second pair wheels **106** that propel the vehicle **100** along a ground surface **108**. In the illustrative embodiment, the vehicle is shown as a bi-directional vehicle; however, it should be appreciated that in other examples, the vehicle may be one of any number of vehicles suitable for use with the ramp assembly **114** described below. A first passenger opening **110** is located between the first pair wheels **104** and the second pair of wheels **106**, and provides access to a passenger for sitting or standing in the vehicle **100**. The passenger opening **110** may be modified to increase the size of the opening **110** to provide access, for example, to a passenger seated in a wheelchair to enter and exit the vehicle **100**. The vehicle **100** may include a conventional floor **112** extending throughout the vehicle **100** to support passengers and other objects traveling in the vehicle **100**.

(23) As shown in FIG. 1, the vehicle **100** may be further modified to include the ramp assembly **114** which provides rolling access of a wheelchair from the ground surface **108** into an interior **116** of the vehicle **100**. Prior to deployment, the ramp assembly **114** is positioned adjacent the opening **110** and beneath the conventional floor **112** of the vehicle **100**. In other examples, prior to deployment, the ramp assembly **114** is positioned partially beneath a modified portion **113** of the conventional floor **112**, and the modified portion **113** is flush with the remainder of the conventional floor **112**. In other examples, the modified portion **113** may be sloped such that at least one (but not all) sides of the modified portion **113** are flush with the conventional floor **112**. When the ramp **120** is described herein as being “flush with the conventional floor” the phrase is meant to cover each illustrative example of a conventional and/or modified-conventional floor herein. Additionally, as used herein “beneath” means directly underneath while “below” means at a lower height but not necessarily underneath.

(24) As shown in FIGS. 2A-B, the ramp assembly **114** includes a frame **118**, a ramp **120** movable relative to the frame **118** and a drive block **122** coupled between the frame **118** and the ramp **120**. The drive block may translate to force sliding for movement of the ramp **120** relative to the frame **118**. As will be described below in greater detail below, the ramp **120** is movable between several positions including a stowed position **124**, a lowered position **126**, and a raised position **128**. As suggested by FIGS. 2A-B, in the stowed position **124**, the ramp **120** is positioned in the interior **116** of the vehicle **100** and beneath the conventional floor **112**. As suggested by FIGS. 3A-B, in the lowered position **126**, the ramp **120** is positioned outside the vehicle **100** and a rear end **132** of the ramp **120** is positioned below the conventional floor **112**. As suggested by FIGS. 5A-B, in the raised position **128**, the ramp **120** is positioned outside the vehicle **100** and the rear end **132** of the ramp **120** is positioned flush with the conventional floor **112** while a front end **131** of the ramp **120** rests on the ground.

(25) Referring now to FIGS. 2A-B, the frame **118** includes a top edge **127**, a bottom edge **129**, a front end **133**, a rear end **134**, and a track system **136**. The rear end **134** of the frame **118** is spaced longitudinally apart from the front end **133**, and the track system **136** extends longitudinally between the front end **133** and the rear end **134** of the frame **118**. The longitudinal direction is indicated by arrow **119**. The track system **136** includes a first-side track **138**, a second-side track **140**, and a frame floor **142** extending between the first-side track **138** and the second-side track **140**. While the frame floor **142** is shown in FIG. 2B, the ramp **120** eclipses the frame floor **142** in FIG. 2A.

(26) An opening **139** is defined between the first-side track **138** and the second-side track **140**. The ramp **120** may pass through the opening **139** as the ramp **120** is moved from the stowed position **124** to the lowered position **126**. As such, the opening **139** is sized to receive the ramp **120** and facilitate passage of the ramp **120** therethrough. In some embodiments, the frame floor **142** supports the ramp **120** as the ramp **120** is moved between the stowed position **124** and the lowered position **126**. It should be understood that in the illustrative embodiment the first-side track **138** and the second-side track **140** are a mirror image of one another and are otherwise identical such that description of a side track herein applies equally to both the first-side track **138** and the second-side

track **140** of the track system **136**.

(27) In some embodiments, the ramp assembly **114** includes a pair of rollers (not shown) coupled to the drive block **122** for rotation relative to the drive block **122**. The rollers are sized and shaped to be positioned within and supported by the first-side track **138**. Rolling movement of the rollers along the first-side track **138** facilitates sliding movement of the ramp **120** relative to the frame **118**.

(28) As shown in FIG. 2B, in some embodiments, the ramp assembly **114** includes an elongated mount **144** fixed to the drive block **122** for sliding movement with the drive block **122** relative to the frame **118** in the longitudinal direction **119**. In such embodiments, the elongated mount **144** extends from the drive block **122** away from the first-side track **138** toward the second-side track **140**. The elongated mount **144** includes a first end **146** coupled to the drive block **122** and a second end **148** coupled to a casing-mount **150**. As shown in FIGS. 2A-B, the casing-mount **150** is coupled to and supports a threaded-casing **152**. In some embodiments, one or more of the threaded-casing **152**, the casing mount **150**, and the elongated mount **144** are formed as a single monolithic component. In other embodiments, each piece may be formed separately and later assembled. In either case, the frame **118** includes a first support bracket **154** and a first bearing **156** each positioned at the front end **134** of the frame **118**.

(29) As shown in FIGS. 2A-B, the first support mount **154** is coupled to the top edge **127** of the frame **118** and extends from the first-side track **138** to the second-side track **140**. The first bearing **156** is coupled to and supported by the support mount **154**. The ramp assembly **114** further includes a threaded-rod **158** extending the front end of the frame **118** to the rear end of the frame **118**. The threaded-rod **158** includes a first end **160** with a portion thereof positioned in the first bearing **156**. The first bearing **156** is configured to support and facilitate rotation of the threaded-rod **158**. The first end **160** of the threaded-rod **158** may be coupled to a rotational power supply (not shown) to cause rotation of the threaded-rod **158**. The threaded-rod **158** may be positioned within and threadingly coupled to the threaded-casing **152**, such that rotation of the thread-rod **158** causes the threaded-casing **152** to translate longitudinally along the threaded-rod **158**. Longitudinal movement of the thread-casing **152** causes sliding movement of the drive block **122**, which is indirectly coupled to the threaded-casing **152** as described above.

(30) As suggested in FIGS. 2A, 3A, and 4A, the frame **118** includes a second support mount **162** and a second bearing **164** each positioned at the front end **133** of the frame **118**. The second support mount **162** is coupled to the top edge **127** of the frame **118** and extends from the first-side track **138** to the second-side track **140**. The second support mount **162** cooperates with the frame floor **142**, the first-side track **138**, and the second-side track **140** to define the opening **139**. The second bearing **164** is coupled to and supported by the second support mount **162**. The threaded-rod **158** includes a second end **166**, and a portion of the second end **166** is positioned in the second bearing **164**. The second bearing **164** is configured to support and facilitate rotation of the threaded-rod **158**, which ultimately facilitates sliding movement of the drive block **122** relative to the frame **118** as described above. It should be appreciated that the drive block **122** may be translated in the longitudinal direction **119** in a number of ways including but not limited to the manner and with the mechanisms described above.

(31) As shown in FIG. 2B, the ramp assembly **114** includes a wheel **168** coupled to the drive block **122** for rotation about a longitudinal axis **170** of the wheel **168**. In the illustrative embodiment, the wheel **168** is a toothed-wheel, meaning that the wheel **168** has a plurality of projections **172** extending away from the center of the wheel **168**. In the illustrative embodiment, the wheel **168** is fixed to an elongated wheel-mount **174** such that the elongated wheel-mount **174** is rotatable with the wheel **168**. In some embodiments, the wheel **168** and the elongated wheel-mount **174** may be formed as a single monolithic component. In other embodiments, each piece may be formed separately and later assembled. In either case, the elongated wheel-mount **174** is coupled to a hinge **176**. As shown in FIG. 2B, the hinge **176** includes a first end **178** coupled to the elongated wheel-

mount **174** and a second end **180** coupled to the rear end **132** of the ramp **120**. The first end **178** of the hinge **176** is configured to pivot relative to the second end **180** of the hinge **176**. As a result, the ramp **120** is pivotally coupled to the wheel **168**. This arrangement also facilitates sliding movement of the ramp **120** relative to the frame **118**. For example, when the hinge **176** is in an opened position as shown in FIG. 2B, the hinge **176** cooperates with the elongated wheel-mount **174** and the wheel **168** to couple the ramp **120** to the drive block **122**. Thus, longitudinal movement of the drive block **122** may push or pull the ramp **120** in the longitudinal direction **119** relative to the frame **118**.

(32) As shown in FIG. 2A, when the ramp **120** is in the stowed position **124**, the wheel **168** is spaced apart from the front end **133** of the frame **118**. As the wheel **168** is moved longitudinally from the rear end **134** of the frame **118** toward the front end **133** of the frame **118**, the ramp **120** is moved from the stowed position **124** shown in FIGS. 2A-B to the lowered position **126** shown in FIGS. 3A-B. As suggested in FIGS. 3A-B, when the ramp **120** is in the lowered position **126**, the wheel **168** is positioned at the front end **133** of the frame **118** and the rear end **132** of the ramp **120** is positioned below the conventional floor **112** (or the modified portion **113**). It should be appreciated that when the ramp assembly **114** is secured in the vehicle **100**, the conventional floor **112** (or the modified portion **113**) is located at a height immediately above the second support mount **162**.

(33) As shown in FIG. 3B, the second support mount **162** includes a toothed-portion **182** having a plurality of projections **184**, and therefore, the support mount **162** may be referred to as a toothed-mount **162**. In some embodiments, the support mount **162** and toothed-portion **182** may be a single monolithic component. In another embodiment, the support mount **162** and the toothed-portion **182** may be formed separately and later assembled, in which case, the toothed-portion **182** may be referred to as a toothed-bar **182**. In either case described above, the projections **184** may be positioned at the front end **133** of the frame **118**.

(34) Referring still to FIG. 3B, in the lowered position **126**, a single projection **172** of the wheel **168** is in contact with a single projection **184** of the support mount **162**. As the drive block **122** continues to slide longitudinally toward the front end **133** of the frame **118**, each of the plurality of projections **172** of the wheel **168** engage with each of the plurality of projections **184** of the second support mount **162** to cause rotation of the wheel **168** about the longitudinal axis **170**. When the wheel **168** and the wheel-mount **174** pivot relative to the drive block **122**, the hinge **176** secured to the drive block **122** moves between an open position (see FIG. 3B) and a folded position (see FIG. 4B) to raise and lower the ramp **120**.

(35) FIGS. 1 and 4A-B show the ramp in the raised position **128**. As shown in FIG. 1, in the raised position **128**, the ramp **120** is positioned outside the motor vehicle **100** and the rear end **132** of the ramp **120** is positioned flush with the conventional floor **112** (or the modified portion **113**). Referring again to FIGS. 4A-B, when the ramp **120** is in the raised position **128**, the wheel **168** is positioned at the front end **133** of the frame **118** and rotated relative to the position of the wheel **168** when the ramp **120** is in the lowered position **126**. In other words, the wheel **168** is rotatable between a first position in which the ramp **120** is in the lowered position **126** (or the stowed position **124**) and a second position in which the ramp **120** is in the raised position **128**. When the wheel **168** is in the first position, the rear end **132** of the ramp **120** is positioned below the top edge **127** of the ramp **120**, and when the wheel **168** is in the second position, the rear end **132** of the ramp **120** is positioned above the top edge **127** of the ramp **120**.

(36) In some embodiments, the wheel **168** is configured to rotate approximately 90 degrees as the ramp **120** is moved between the lowered position **126** and the raised position **128**. As the wheel **168** is rotated 90 degrees, the first end **178** of the hinge **176** moves between: (i) a first position in which the first end **178** of the hinge **176** extends in a plane generally parallel to the frame floor **142**, and (ii) a second position in which the first end **178** of the hinge **176** extends in a plane that is generally perpendicular to the frame floor **142**.

(37) While the description above has described the toothed-wheel **168** coupled to the drive block

122 and the toothed-bar **182** as coupled to or formed in the second support mount **162**, it should be appreciated that the location of the of toothed-wheel **168** and the toothed-bar **182** may be reversed. For example, in illustrative embodiments, the toothed-bar **182** may be fixed to the drive block **122** for sliding movement therewith and the toothed-wheel **168** may be pivotally coupled to the frame **118** at a fixed location on the frame **118**. As the drive block **122** slides longitudinally toward the front end **133** of the frame **118**, the toothed-bar **182** slides into contact with the toothed-wheel **168**. As the drive block **122** continues to slide longitudinally toward the front end **133** of the frame **118**, each of the plurality of projections **184** of the toothed-bar **182** engage with each of the plurality of projections **172** of the toothed-wheel **168** to cause rotation of the toothed-wheel **168**. In such an embodiment, rotation of the toothed-wheel **168** causes the hinge **176** to move between an open and a folded position to raise and lower the ramp **120** relative to the frame **118**.

(38) In the illustrative embodiment shown in FIG. 5, the vehicle **200** includes a unibody construction, but other vehicles having a frame on body construction are also included in the present disclosure. Consequently, the use of vehicle herein includes all types and kinds of vehicles with a body on frame construction, a unibody construction, or other constructions. In addition, while the passenger van **200** is illustrated in FIG. 5, the present disclosure is directed to all passenger vehicles carrying one or more passengers including, but not limited to, a bus, motor coach, sport-utility vehicle, truck, taxi, ambulance, or passenger car.

(39) The vehicle **200** includes a frame or chassis **202** coupled to a first pair of wheels **204** and a second pair wheels **206** that propel the vehicle **200** along a ground surface **208**. In the illustrative embodiment, the vehicle is shown as a bi-directional vehicle; however, it should be appreciated that in other examples, the vehicle may be one of any number of vehicles suitable for use with a ramp assembly **214** described below. A first passenger opening **210** is located between the first pair wheels **204** and the second pair of wheels **206**, and provides access to a passenger for sitting or standing in the vehicle **200**. The passenger opening **210** may be modified to increase the size of the opening **210** to provide access, for example, to a passenger seated in a wheelchair to enter and exit the vehicle **200**. The vehicle **200** may include a conventional floor **212** extending throughout the vehicle **200** to support passengers and other objects traveling in the vehicle **200**.

(40) As shown in FIG. 5, the vehicle **200** may be further modified to include the ramp assembly **214** which provides rolling access of a wheelchair from the ground surface **208** into an interior **216** of the vehicle **200**. Prior to deployment, the ramp assembly **214** is positioned adjacent the opening **210** and beneath the conventional floor **212** of the vehicle **200**. In other examples, prior to deployment, the ramp assembly **214** is positioned partially beneath a modified portion **213** of the conventional floor **212**, and the modified portion **213** is flush with the remainder of the conventional floor **212**. In other examples, the modified portion **213** may be sloped such that at least one side of the modified portion **113** is flush with the conventional floor **212**. In reference to the ramp **120**, the phrase “flush with the conventional floor” is meant to cover each illustrative example of a conventional and/or modified-conventional floor described herein. Additionally, as used herein “beneath” means directly underneath while “below” means at a lower height but not necessarily underneath.

(41) As shown in FIGS. 6A-B, the ramp assembly **214** includes a frame **218**, a ramp **220** movable relative to the frame **218**, and a drive block **222** coupled between the frame **218** and the ramp **220**. The drive block may translate (i.e., slide) to force the ramp **220** to slide relative to the frame **218**. As will be described below in greater detail below, the ramp **220** is movable between several positions including a stowed position **224**, a lowered position **226**, and a raised position **228**. As suggested by FIGS. 6A-B, in the stowed position **224**, the ramp **220** is positioned in the interior **216** the vehicle **200** and beneath the conventional floor **212**. As suggested by FIGS. 7A-B, in the lowered position **226**, the ramp **220** is positioned outside the vehicle **200** and a rear end **232** of the ramp **220** is positioned below the conventional floor **212**. As suggested by FIGS. 8A-B, in the raised position **228**, the ramp **220** is positioned outside the vehicle **200** and the rear end **232** of the

ramp **220** is positioned flush with the conventional floor **212** while a front end **231** of the ramp **220** rests on the ground **208**.

(42) Referring now to FIGS. **6A-B**, the frame **218** includes a top edge **227**, a bottom edge **229**, a front end **233**, a rear end **234**, and a track system **236**. The rear end **234** of the frame **218** is spaced longitudinally apart from the front end **233**, and the track system **236** extends longitudinally between the front end **233** and the rear end **234** of the frame **218**. The longitudinal direction is indicated by arrow **219**. The track system **236** includes a first-side track **238**, a second-side track **240**, and a frame floor **242** extending between the first-side track **238** and the second-side track **240**. The frame floor **242** is shown in FIG. **6B**, and the ramp **220** eclipses the frame floor **242** in FIG. **6A**.

(43) An opening **239** is defined at the front end **233** of the frame **218** between the first-side track **238** and the second-side track **240**. The ramp **220** may pass through the opening **239** as the ramp **220** is moved from the stowed position **224** to the lowered position **226**. As such, the opening **239** is sized to receive the ramp **220** and facilitate passage of the ramp **220** therethrough. In some embodiments, the frame floor **242** supports the ramp **220** as the ramp **220** is moved between the stowed position **224** and the lowered position **226**. It should be understood that in the illustrative embodiment the first-side track **238** and the second-side track **240** are a mirror image of one another and are otherwise identical such that description of a side track herein applies equally to both the first-side track **238** and the second-side track **240** of the track system **236**.

(44) In some embodiments, the ramp assembly **214** includes rollers **215** sized and shaped to be positioned within and supported by the first-side track **238**. The ramp assembly **214** also includes a spacer block **223** coupled between the drive block **222** and the rollers **215**. The rollers **215** are pivotally coupled to the spacer block for rotational movement relative to the spacer block **223** and the drive block **222**. Rolling movement of the rollers **215** along the first-side track **238** facilitates sliding movement of the ramp **220** relative to the frame **218**.

(45) As shown in FIG. **6B**, in some embodiments, the ramp assembly **214** includes an elongated mount **244** fixed to the drive block **222** for sliding movement with the drive block **222** relative to the frame **218** in the longitudinal direction **219**. In such embodiments, the elongated mount **244** is coupled to and extends from the drive block **222** away from the first-side track **238** toward the second-side track **240**. The elongated mount **244** includes a first portion **246** coupled to the drive block **222** and a second portion **248** coupled to a first casing mount **250**. As shown in FIGS. **6A-B**, the first casing mount **250** is coupled to and supports a first threaded casing **252**. In some embodiments, one or more of the first threaded casing **252**, the casing mount **250**, and the elongated mount **244** are formed as a single monolithic component. In other embodiments, each piece may be formed separately and later assembled. The ramp assembly **214** also includes a first support bracket **254** including a first bearing **256** formed therein.

(46) As shown in FIG. **6A** many components of the ramp assembly **214** have a mirror image counterpart. For example, the ramp assembly **214** includes a second casing mount **251** coupled to and supporting a second threaded casing. Additionally, the ramp assembly **214** includes a second support bracket **259** including a second bearing **261** formed therein. It should be appreciated that the description of any mirror image component applies equally to its counterpart. A cross bar **253** is positioned above and coupled to the first support bracket **254** and the second support bracket **259**. The cross bar **253** is fixed to the frame **218**, positioned at the top edge **227** of the frame **218**, and extends transversely across the frame **218** perpendicular to the longitudinal direction **219**. The cross bar **253** is one of a plurality of cross bars configured to support the conventional floor **212** or modified portion thereof **213** above the internal components of the ramp assembly **214**.

(47) Referring still to FIGS. **6A-B**, the ramp assembly **214** further includes a first threaded rod **258** extending from the front end of the frame **218** to the rear end of the frame **218**. The first threaded rod **258** includes a first end **260** with a portion thereof positioned in the first bearing **256**. The first bearing **256** is configured to support and facilitate rotation of the first threaded-rod **258**. The first

end **260** of the first threaded rod **258** may be coupled to a rotational power supply (not shown) to cause rotation of the threaded rod **258**. The first threaded rod **258** may be positioned within and threadingly coupled to the first threaded casing **252**, such that rotation of the first threaded rod **258** causes the first threaded casing **252** to translate longitudinally along the first threaded rod **258**. Longitudinal movement of the first threaded casing **252** causes sliding movement of the drive block **222**, which is indirectly coupled to the first threaded casing **252** as described above.

(48) The ramp assembly **214** further includes a second threaded rod **263** extending the front end **233** of the frame **218** to the rear end **234** of the frame **218**. The second threaded rod **263** includes a first end with a portion thereof positioned in the second bearing **261**. The second bearing **261** is configured to support and facilitate rotation of the second threaded rod **263**. The first end of the second threaded rod **263** may be coupled to a rotational power supply (not shown) to cause rotation of the second threaded rod **263**. The second threaded rod **263** may be positioned within and threadingly coupled to a second threaded casing, such that rotation of the second threaded rod **263** causes the second threaded casing to translate longitudinally along the second threaded rod **263**. The first and second threaded rods **258**, **263** may be coupled to the same rotational power supply and are rotated at the same speed to move their respective threaded casings and drive blocks at the same speed longitudinally along the track system **236**, which ultimately moves the ramp **220** in a smooth fashion relative to the frame **218**.

(49) As shown in FIGS. **6A** and **7A-B**, the frame **218** includes a second cross bar **265** positioned at the front end **233** of the frame **218**. Like the first cross bar **253**, the second cross bar **265** is fixed to the frame **218**, positioned at the top edge **227** of the frame **218**, and extends transversely across the frame **218**. The second cross bar **265** cooperates with the frame floor **242**, the first-side track **238**, and the second-side track **240** to define the opening **239**.

(50) The second cross bar **265** includes a first support mount **267** having a bearing **269** formed therein and a second support mount **271** having a bearing **273** formed therein. The first threaded rod **258** includes a second end positioned in the bearing **269**. The bearing **269** is configured to support and facilitate rotation of the first threaded rod **258**, which ultimately causes sliding movement of the drive block **222** relative to the frame **218** as described above. The second threaded rod **263** includes a second end positioned in the bearing **273**. The bearing **273** is configured to support and facilitate rotation of the second threaded rod **263**, which ultimately causes sliding movement of the respective drive block, as described above. Sliding movement of the drive blocks causes sliding movement of the ramp **220** relative to the frame **218**, as described above.

(51) As shown in FIG. **6B**, the ramp assembly **214** includes a wheel **268** coupled to the drive block **222** for rotation about a longitudinal axis **270** of the wheel **268**. In the illustrative embodiment, the wheel **268** is a toothed wheel, meaning that the wheel **268** has a plurality of projections **272** extending away from the center of the wheel **268**.

(52) As shown in FIG. **7C**, the plurality of projections **272** includes a keystone projection **275**, a first plurality of subordinate projections **277** positioned clockwise of the keystone projection **275**, and second plurality of subordinate projections **279** positioned counter-clockwise of the keystone projection **275**. Each subordinate projection is spaced apart a first distance **291** from an adjacent subordinate projection. The first and second pluralities of subordinate projections **277**, **279** each include a first projection **281** positioned nearest to the keystone projection **275**. Each first projection **281** is spaced apart from the keystone projection **275** a second distance **293** that is greater than the first distance **291**. As shown in FIG. **6A**, the ramp assembly **214** includes two wheels **268**. In the illustrative embodiment, the two wheels **268** are identical to one another and not mirror image components. The existence of two pluralities of subordinate projections, e.g., one on each side of the keystone projection **275**, allows the same wheel **268** to be positioned on both sides of the ramp assembly **214**. Thus, only one configuration of a toothed wheel is required to be manufactured to assemble the ramp assembly **214**.

(53) Referring again to FIG. 6B, in the illustrative embodiment, the wheel 268 is fixed to an elongated wheel mount 274. The elongated wheel mount 274 is rotatable with the wheel 268. In some embodiments, the wheel 268 and the elongated wheel mount 274 may be formed as a single monolithic component. In other embodiments, each piece may be formed separately and later assembled. In either case, the elongated wheel mount 274 is coupled to a hinge 276. As shown in FIG. 6B, the hinge 276 includes a first end 278 coupled to the elongated wheel mount 274 and a second end 280 coupled to the rear end 232 of the ramp 220. The first end 278 of the hinge 276 is configured to pivot relative to the second end 280 of the hinge 276. As a result, the ramp 220 is pivotally coupled to the wheel 268. This arrangement also facilitates sliding movement of the ramp 220 relative to the frame 218. For example, when the hinge 276 is in an opened position as shown in FIG. 6B, the hinge 276 cooperates with the elongated wheel mount 274 and the wheel 268 to couple the ramp 220 to the drive block 222. Thus, longitudinal movement of the drive block 222 may urge (i.e., push or pull) the ramp 220 in the longitudinal direction 219 relative to the frame 218.

(54) As shown in FIG. 6A, when the ramp 220 is in the stowed position 224, the wheel 268 is spaced apart from the front end 233 of the frame 218. As the wheel 268 is moved longitudinally from the rear end 234 of the frame 218 toward the front end 233 of the frame 218, the ramp 220 is moved from the stowed position 224 shown in FIGS. 6A-B to the lowered position 226 shown in FIGS. 7A-B. As suggested in FIGS. 7A-B, when the ramp 220 is in the lowered position 226, the wheel 268 is positioned at the front end 233 of the frame 218, and the rear end 232 of the ramp 220 is positioned below the conventional floor 212 or the modified portion thereof 213.

(55) FIG. 7B shows the second cross bar 265; however, a portion of the cross bar 265 has been cut away to show a toothed portion 282 formed in the cross bar 265. The toothed portion 282 includes a plurality of projections 284 (see FIG. 6C). In some embodiments, the toothed portion 282 and the cross bar 265 may be a single monolithic component. In some embodiments, the toothed portion 282 may be formed in a bar or other component that is distinct from and merely coupled to the cross bar 265. At times, the toothed portion 282 may be referred to as a toothed bar.

(56) In the illustrative embodiment, the toothed portion 282 includes a primary projection 283 and a plurality of secondary projections 285. When the tooth portion 282 is formed in the cross bar 265 and the cross bar 265 is coupled to the frame 218 as illustratively shown in FIG. 7B, the plurality of secondary projections 285 are positioned forward of the primary projection 283. The primary projection 283 has a width 295 that is substantially equal to the second distance 293 defined between the keystone projection 275 and the first subordinate projections 281.

(57) As shown in FIG. 7B, when the ramp 220 is in the lowered position 226, the keystone projection 275 of the wheel 268 is in contact with the primary projection 283 of the cross bar 265. As the drive block 222 continues to slide longitudinally toward the front end 233 of the frame 218, the plurality of subordinate projections 279 of the wheel 268 engage with each of the plurality of secondary projections 285 of the cross bar 265 to cause rotation of the wheel 268 about the longitudinal axis 270. When the wheel 268 and the wheel mount 274 pivot relative to the drive block 222, the hinge 276 moves between an opened position (see FIG. 7B) and a folded position (see FIG. 8B) to raise and lower the ramp 220.

(58) Referring still to FIG. 7B, the ramp assembly 214 further includes a roller 287 supported above the frame floor 242 by the first support mount 267. The ramp assembly 214 also includes a pad 289 coupled to the first end 278 of the hinge 276 and aligned with the roller 287. When the ramp 220 is in the lowered position 226 (or stowed position 224), an exposed surface of the pad 289 faces away from the frame floor 242. As best shown in FIG. 6A, it should be appreciated that the ramp assembly 214 includes a second roller 297 supported above the frame floor 242 by the second support mount 271. The ramp assembly 214 also includes a second pad 299 coupled to the first end 278 of the hinge 276 and aligned with the roller 297. When the ramp 220 is in the lowered position 226 (or stowed position 224), an exposed surface of the pad 299 faces away from the

frame floor **242**.

(59) As the ramp **220** is moved from the stowed position **224** to the lowered position **226** the ramp **220** becomes cantilevered, extending beyond the forward end **233** of the frame **218**. Therefore, gravity has a tendency to force the front end **231** of the ramp **220** downward, thus, the rearward end **232** of the ramp **220** is forced upward. To counteract these forces, the pads **289**, **299** contact the rollers **287**, **297** to limit upward and downward movement of the ramp **220** as the ramp is moved from the stowed position **224** to the lowered position **226**.

(60) As shown in FIG. 7B, the ramp **220** further includes support mechanism **245** coupled between the elongated mount **244** and the elongated wheel mount **274**. As shown in the illustrative embodiment, the support mechanism **245** may include, for example, a gas spring **247**. In any case, the support mechanism **245** is fixed at a first end to the elongated mount **244** and at a second end to the elongated wheel mount **274** to further prevent upward and downward movement of the ramp **220** when the ramp **220** is moved from the stowed position **224** to the lowered position **226**. In the illustrative embodiment, the gas spring **247** is positioned in a support mechanism housing **249**. Pins **251** extend through ends of the gas spring **247** and ends of housing **249** to couple the gas spring **247** to the housing **249**.

(61) FIGS. 1 and 8A-B show the ramp in the raised position **228**. As shown in FIG. 1, in the raised position **228**, the ramp **220** is positioned outside the motor vehicle **200** and the rear end **232** of the ramp **220** is positioned flush with the conventional floor **212** or the modified portion thereof **213**. Referring again to FIGS. 8A-B, when the ramp **220** is in the raised position **228**, the wheel **268** is positioned at the front end **233** of the frame **218** and rotated relative to the position of the wheel **268** when the ramp **220** is in the lowered position **226**. In other words, the wheel **268** is rotatable between a first position in which the ramp **220** is in the lowered position **226** (or the stowed position **224**) and a second position in which the ramp **220** is in the raised position **228**. When the wheel **268** is in the first position, the rear end **232** of the ramp **220** is positioned below the top edge **227** of the frame **218**, and when the wheel **268** is in the second position, the rear end **232** of the ramp **220** is positioned above the top edge **227** of the frame **218** (flush with the conventional floor **212**).

(62) In some embodiments, the wheel **268** is configured to rotate approximately 90 degrees as the ramp **220** is moved between the lowered position **226** and the raised position **228**. As the wheel **268** is rotated 90 degrees, the first end **278** of the hinge **276** moves between: (i) a first position in which the first end **278** of the hinge **276** extends in a plane substantially parallel to the frame floor **242**, and (ii) a second position in which the first end **278** of the hinge **276** extends in a plane that is substantially perpendicular to the frame floor **242**.

(63) While the description above has described the toothed wheel **268** as coupled to the drive block **222** and the toothed portion **282** as formed in or coupled to the cross bar **265**, it should be appreciated that the location of the of toothed wheel **268** and the toothed portion **282** may be reversed. For example, in illustrative embodiments, the toothed portion **282** may be formed in or otherwise fixed to the drive block **222** for sliding movement therewith, and the toothed wheel **268** may be pivotally coupled to the frame **218** and fixed in a location relative to the frame **218**. As the drive block **222** slides longitudinally toward the front end **233** of the frame **218**, the toothed portion **282** slides into contact with the toothed wheel **268**. As the drive block **222** continues to slide longitudinally toward the front end **233** of the frame **218**, each of the projections **284** of the toothed portion **282** engage with each of the plurality of projections **272** of the toothed wheel **268** to cause rotation of the toothed wheel **268**. The toothed wheel **268** and the toothed bar **282** are shown generally in FIG. 6C. In such an embodiment, rotation of the toothed wheel **268** causes raising and lowering the ramp **220** relative to the frame **218**, as described above.

(64) The term “cog” is meant to include both a wheel having a series of projections and a bar having a series of projections. As used herein a cog is a wheel or bar that transfers motion by engaging with projections on another wheel or bar. In the illustrative embodiments shown herein, a

first cog is coupled to the frame **218** at a fixed location on the frame **218**, and a second cog is coupled to the ramp **220** for sliding movement with the ramp **220**. The wheel **268** is rotatable relative to the component to which it is coupled. This facilitates movement of the ramp **220** between the lowered position **226** and the raised position **228**.

(65) Referring now to FIG. **8A**, a first side rail **302** and a second side rail **304** of the ramp assembly **214** are shown. In FIG. **8A**, the side rails **302**, **304** have been lifted out of their respective side rail housings **308**, **310**. As shown FIG. **7A**, the side rails **302**, **304** are in a collapse position within the housings **308**, **310** when the ramp **220** is in the lowered position **226** (or the stowed position **224**). As shown in FIG. **8A**, the side rails **302**, **304** are in a lifted position when the ramp **220** is in the raised position **228**. As the ramp **220** moves from the lowered position **226** to the raised position **228**, the spacer block **223** and the cross bar **265** contact a lever **306** of the side rail **302** to cause rotation of the lever **306** relative to the ramp **220**. Contact by the spacer block **223** and the cross bar **265** forces the side rail **302** to move to the lifted position, as described in greater detail below.

(66) Referring now to FIGS. **9** and **10**, the side rail **302** is shown; however, it should be appreciated that the side rails **302** and **304** are identical such that any description of the side rail **302** applies equally to the side rail **304**. FIG. **9** shows the side rail **302** in the collapse position, and FIG. **10** shows the side rail **302** in the lifted position.

(67) As shown in FIGS. **9** and **10**, the ramp assembly **214** includes a support arm **312** coupled to the drive block **222** for sliding movement therewith. The support arm **312** includes a pocket **314**. The ramp **220** includes a weight-bearing pin **316** coupled to the ramp **220** and configured to rest in the pocket **314** of the support arm **312**. The weight-bearing pin **316** may also travel longitudinally within the pocket **314**. When the ramp **220** is in the lowered position **226** (and the stowed position **224**) the weight-bearing pin **316** is positioned at a front end of the pocket **314**. When the ramp **220** is in the raised position **228**, the weight-bearing pin **316** is positioned at a rearward end of the pocket **314**. As the ramp **220** is moved from the stowed position **224** to the lowered position **226** the ramp **220** becomes cantilevered, extending beyond the forward end **233** of the frame **218**. Therefore, gravity has a tendency to force the front end **231** of the ramp **220** downward, thus, the rearward end **232** of the ramp **220** is forced upward. To counteract these forces, the support arm **312** is configured to support the weight-bearing pin **316** (and the ramp **220**) above the ground **208**.

(68) As shown in FIGS. **9** and **10**, the side rail **302** includes the lever **306**, a linkage bar **318**, and a runner **320**. The lever **306** includes a first end **322** pivotally coupled to the runner **320** and a second end or free end **324** positioned opposite the first end **322**. The lever **306** further includes a ramped portion **326** extending upwardly and away from the free end **324**. The lever **306** is pivotally coupled to the ramp **220** at a rotational axis **328** of the lever **306**. The rotational axis **328** is positioned between the first end **322** and the second end **324** of the lever **306**. When the ramp **220** moves from the lowered position **226** to the raised position **228**, the spacer block **223** and the cross bar **265** contact the free end **324** and the ramped portion **326** to cause rotation of the lever **306**. As the lever **306** rotates, the lever **306** pulls the runner **320** out from the housing **308** and moves the side rail **302** to the lifted position. In the illustrative embodiment, the linkage bar **318** is an L-shaped bar having a first end **330** pivotally coupled to the ramp **220** and a second end **332** pivotally coupled to the runner **320**. When the runner **320** is lifted out from the housing **308**, the runner **320** pulls the linkage bar **318** out from the housing **308** as well. The linkage bar **318** rotates about its first end **330** and second end **332** as the side rail **302** moves to the lifted position.

(69) While exemplary embodiments incorporating the principles of the present disclosure have been disclosed herein, the present disclosure is not limited to the disclosed embodiments. Instead, this application is intended to cover any variations, uses, or adaptations of the disclosure using its general principles. For instance, the present disclosure is not limited to the modification of a purchased OEM vehicle, but can be incorporated into the OEM vehicle when manufactured. Further, this application is intended to cover such departures from the present disclosure as come

within known or customary practice in the art to which this disclosure pertains and which fall within the limits of the appended claims.

Claims

1. A ramp assembly to accommodate a wheel-chaired passenger to enter or exit a motorized vehicle having a floor, the ramp assembly comprising: a ramp coupled to a drive block by a hinge wherein the drive block is configured to move the ramp from a stowed position to a deployed position; the hinge having a first end rotatable with and about the same axis as a wheel and a second end pivotally coupled to the ramp, wherein rotation of the wheel causes the ramp to raise relative to the wheel from a lowered position to a use position and lower relative to the wheel from the use position to the lowered position.
2. The ramp assembly of claim 1, wherein the ramp has a partially deployed position between the stowed position and the deployed position.
3. The ramp assembly of claim 2, wherein in the partially deployed position the ramp is also in the lowered position.
4. The ramp assembly of claim 3, wherein in the deployed position the ramp is also in the use position.
5. The ramp assembly of claim 4, wherein the wheel rotates to raise the ramp relative to the wheel from the lowered position to the use position as the ramp moves from the partially deployed position to the deployed position.
6. The ramp assembly of claim 4, wherein the drive block is configured to move in a generally horizontal plane relative to the vehicle as it moves the ramp between the stowed position and the deployed positions.
7. The ramp assembly of claim 6, wherein the first end and the second end of the hinge are disposed in the generally horizontal plane when the ramp is between the stowed position and the partially deployed position.
8. The ramp assembly of claim 7, wherein the first and second ends of the hinge are disposed at an angle relative to the generally horizontal plane when the ramp is in the use position.
9. The ramp assembly of claim 4, wherein the wheel is disengaged from a frame of the ramp assembly when the ramp is between the stowed and partially deployed position, and engages with the frame as the drive block moves the ramp between the partially deployed position and the deployed position.
10. A motorized vehicle comprising a floor and a ramp assembly, the ramp assembly having: a ramp coupled to a drive block by a hinge wherein the drive block is configured to move the ramp from a stowed position to a deployed position; the hinge having a first end rotatable with and about the same axis as a wheel and a second end pivotally coupled to the ramp, wherein rotation of the wheel causes the ramp to raise and lower relative to the wheel from a lowered position to a use position and lower relative to the wheel from the use position to the lowered position.
11. The motorized vehicle of claim 10, wherein the ramp is disposed below the floor in the stowed position and wherein a rear end of the ramp is generally flush with the floor in the deployed position.
12. The motorized vehicle of claim 11, wherein the ramp has a partially deployed position between the stowed position and the deployed position.
13. The motorized vehicle of claim 12, wherein in the partially deployed position the ramp is also in the lowered position.
14. The motorized vehicle of claim 13, wherein in the deployed position the ramp is also in the use position.
15. The motorized vehicle of claim 14, wherein the wheel rotates to raise the ramp relative to the wheel from the lowered position to the use position as the ramp moves from the partially deployed

position to the deployed position.

16. The motorized vehicle of claim 15, wherein the drive block is configured to move in a generally horizontal plane relative to the vehicle as it moves the ramp between the stowed position and the deployed positions.

17. The motorized vehicle of claim 16, wherein the first end and the second end of the hinge are disposed in the generally horizontal plane when the ramp is between the stowed position and the partially deployed position.

18. The motorized vehicle of claim 17, wherein the first and second ends of the hinge are disposed at an angle relative to the generally horizontal plane when the ramp is in the use position.

19. The motorized vehicle of claim 18, wherein the wheel is disengaged from a frame of the ramp assembly when the ramp is between the stowed and partially deployed position, and engages with the frame as the drive block moves the ramp between the partially deployed position and the deployed position.

20. A method for accommodating a wheel-chaired passenger to enter or exit the motorized vehicle of claim 14, the method comprising operating the drive block to move the ramp from the stowed position to the deployed position, wherein: the wheel engages a frame of the ramp assembly when the ramp reaches the partially deployed position; and the wheel rotates by virtue of engagement with the ramp assembly as the ramp moves between the partially deployed position to the deployed position, wherein the ramp moves from the lowered position to the use position.
